

Liquid Extraction

Pre-Lab Plan
Unit Operations Lab 3
February 18, 2020

Group #1, Tuesday, AM section

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Experimental Objective (5pts)

The emphasis of the experiment is to study the characteristics of liquid-liquid extraction using a three stage mixer-settler system containing water as the solvent and 0.5 wt% acetic acid in white oil. The specific objectives are to measure the concentration of acetic acid as a function of stage number and agitator speed for the two phases, use a McCabe-Thiele diagram to obtain the number of theoretical stages, to determine the Murphree efficiencies for the stages, and finally to calculate an overall efficiency.

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Commented [T3]: Objective*, not "emphasis"

Commented [T4]: With different agitator speed

2. Experimental

The materials, supplies, and the apparatus of this experiment are introduced below. As pictured, there are three different stages that will contain both a mixture of white oil/acetic acid and water in order to measure the liquid-liquid extraction of acetic acid from oil to water.

1. Apparatus (5 pts)

Figure 1: Apparatus used during the Liquid-Liquid Extraction experiment.

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Figure 2: Back Side of the apparatus used during the Liquid-Liquid Extraction experiment.



Figure 3: Pump used during the Liquid-Liquid Extraction experiment

Table 1. List of apparatus in the Liquid Extraction unit with their description

1	Tanks	The white oil and acetic acid mixture will be in the feed tank at the beginning of the experiment. At the end of the experiment, the white oil and small amount of acetic acid will be in the product tank.
2	Agitator	There is an agitator in each of the three stages, to ensure that the stage is mixed well.
3	Mixer-settler Stages	The bins are where the mixing of the liquid-liquid extraction occurs. There are three, as there are three stages in this experiment.
4	Switch	This switch is used to power the three agitators.
5	Switch (pump)	This switch is used to power the pump.
6	Recycle Valve	This recycle valve is used to recycle the water to create a closed circuit.
7	Drain	There are a total of 2 drains that are present on each bin, for a total of 6 drains. The drains will not be used in the experiment due to it being a closed system.
8	Water Valve	This is used to allow the water to enter the system. This is used to control the water that is being taken from the house supply.
9	Water Rotameter	This is used in order to adjust flow rate of water that is being put into the system
10	White Oil Rotameter	This is used in order to adjust the flow rate of white oil that is being pumped into the system.
11	Pump	This is used in order to pump the white oil into the three different stages.
12	Agitator Dials	These dials are used to vary the speed of the agitator. There is one on each agitator, for a total of 3.
13	White Oil Flow Valve	This is used to allow the white oil mixture to enter the system. This is also used to control the white oil that is being taken from the feed tank.
14	Product Return Pipe	This pipe returns the product to the feed tank. This will only be operated by the instructor.

Commented [T7]: Which tank is the oil feed tank? And which one is the product tank?

2. Materials and Supplies (5 pts)

Table 2: Description of materials and supplies used during this experiment.

1	TG 55 White mineral oil	The mineral oil will be used as the solvent for initial acetic acid
2	Glacial acetic acid	Acetic acid will be extracted from the white mineral oil.
3	Water	Water will be the solvent used to separate the acetic acid from the mineral oil.
4	0.05 M NaOH	Used to determine the acetic acid concentration, a standardized solution of 0.05 M NaOH will be used to titrate the acetic acid mixture.
5	Phenolphthalein	This will be the indicator used to determine the concentration of the acetic acid during titrations.
6	50 mL Burette w/Clamp	This will be used to hold the NaOH solution to perform titrations.
7	Serologic Pipet	The serologic pipet will be used to transfer the sample into a beaker for future titration. The size of the serologic pipet is 25 mL in 2/10 intervals.
8	Beaker/Erlenmeyer Flask	This will be used to store the samples to be titrated.
9	Rubber bulb	The rubber bulb will be used to suck up the sample into the serologic pipet.
10	Stir bar	This will be used to stir the sample mixture during titration.
11	Chemical resistant gloves	This is for protection because glacial acetic acid will be used.

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Commented [T9]: And stir plate, and you'll need parafilm

3. Experimental Plan (10 pts)

Independent variable: The independent variable for this experiment is the agitator speed.

Dependent variable: The dependent variables for this experiment are the efficiencies, number of theoretical stages, and the acetic acid concentration.

1. Ensure the White Oil is in the feed tank and at a volume of 25 gallons. If one of these cases are not true, ask the TA/Instructor for assistance.
2. Prepare a 25-gallon solution of White Oil and Acetic Acid that is 0.5 wt% Acetic Acid.
 - a. Perform an initial titration of a sample of the White Oil/Acetic Acid against a 0.05 N NaOH
 - b. Calculate the concentration of Acetic Acid and add the necessary amount in order to reach 0.5 wt%
3. Mix the acetic acid and the White oil in the feed tank by using the centrifugal pump in the closed circuit.
 - i. A closed circuit is a system where mass is conserved within the boundaries of the system. This means that the oil will be recycled back to the feed and will not be removed from the system.
 - b. Take a sample of the mixture to determine the composition of acetic acid within the white oil.
4. Water, the solvent circuit, will be used in the mixer-settler unit in order to reach steady levels for each stage of the experiment.
 - a. Open the water flow valves until the rotameter shows a flowrate of 0.67 GPM.
 - b. Ensure all drain valves are closed.
5. Adjust the speed of the three agitators to 400 rpm each.
6. Once the water circuit has been established, the White Oil feed will be fed into the system.
 - a. The flow of the oil can then be adjusted so that the solvent is being pumped at twice the speed of the feed.
 - i. The White Oil flowrate of 0.34 GPM
 - ii. The flows will be adjusted by trial and error in order to reach steady state within the system.
7. The flowrate of the white oil should be chosen so that three runs can be conducted with the 25 gallons of mixture that is supplied.
8. Samples of White oil and water phases will be withdrawn from each stage and the composition of acetic acid within them will then be determined by performing a titration with 0.05 NaOH
9. The experiment will then be performed three times with different agitator speeds while the flow rate is kept constant.
 - a. The agitator speeds chosen for the experiment where 400 RPM, 600 RPM, and 800 RPM .

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Commented [T11]: And the initial concentration of acetic acid in oil

Commented [T12]: How much acetic acid? In ml?

Project Safety Evaluation (10 pts)

Project Title: Liquid Extraction

Date: 2/18/2020

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	Yes/No	
Potential electrical hazards?	Yes	The electric motors, wires, and electrical box can pose shock hazards if they come in contact with water.
Potential mechanical hazards?	Yes	The rotating impellers can create pinch points if body parts come in contact with them while they are rotating. The mineral oil can make the floor slippery increasing possibility of a fall hazard.
Potential pressure hazards?	No	
Potential temperature hazards?	Yes	The electrical motors can get hot, take caution when touching the motors when they are on or have recently turned off.
Hazardous/toxic chemicals involved?	Yes	Sodium hydroxide, glacial acetic acid and phenolphthalein will be used in this experiment. NaOH and glacial acetic acid could cause severe burns of eyes, skin, and mucous membranes. Phenolphthalein is carcinogenic and an eye irritant.
Gloves required?	Yes	Latex/nitrile gloves and acid resistant gloves for handling glacial acetic acid.
MSDS reviewed by all team members?	Yes	Phenolphthalein CAS# 77-09-8 NaOH CAS# 1310-73-2 TG55 White Mineral Oil CAS# 8042-47-5 Glacial Acetic Acid CAS# 64-19-7
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	40-50 °C	Impeller motors, don't touch the motors, only the dials
Pressure (psig)	1 atm	Experiment is done at atmospheric pressure
Safety Controls	yes/no	Description
Water Inlet Valve	Yes	Main shutoff valve for the inlet water
Main Electrical Switch	Yes	Main shutoff switch for all the electrical equipment for the experiment
Main Pump Switch	Yes	Main switch for the power to the pumps
Fume Hood	Yes	Safe removal of harmful glacial acetic acid vapors from the workspace

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I have read relevant background material for the Unit Operations Laboratory entitled: "Liquid-liquid Extraction" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

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